A Diagnostic Evaluation of CTGAN generated financial data across Fidelity, Utility, and Fairness

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Background and Motivation

- Machine Learning (ML) for decision making in high-stakes domains like finance for tasks like credit risk assessment and fraud detection.
- Access to high quality and representative data is limited[1].
- Synthetic data generation as an alternative for real data.
- CTGAN[2] is known for generating realistic synthetic tabular data.
- Synthetic data should be evaluated beyond fidelity, including downstream utility and fairness.
- Observe the impact of hyperparameter choices on fidelity, utility, and fairness in financial synthetic data generation.

Research Questions

How do CTGAN hyperparameter configurations affect the tradeoff between fidelity, utility, and fairness in synthetic financial data generation?

- SRQ1: Fidelity: Is synthetic data statistically similar to the real data?
- SRQ2: Utility: Does synthetic data support downstream prediction tasks as effectively as real data?
- SRQ3: Fairness: does synthetic data preserve, reduce, exacerbate fairness disparities present in the real data?

Methodology

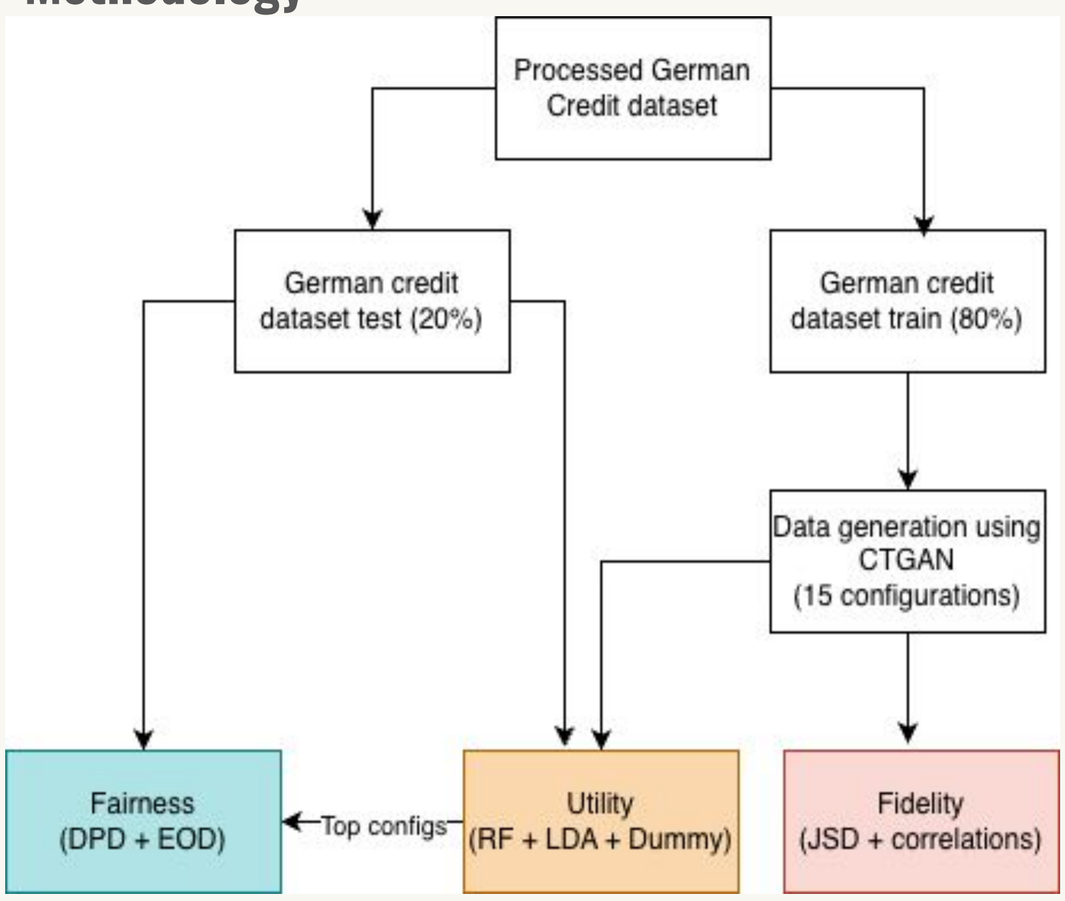


Figure 1: Experimental setup

- German Credit dataset[3]
  - Sensitive attribute: age (<25 or ≥25).
- 15 configurations ⇒ each configuration = one hyperparameter varied at a time.
  - Synthetic data generated using SDV*
- **T**rain on **S**ynthetic, **T**est on **R**eal (TSTR) and **T**rain on **R**eal **T**est on **R**eal (TRTR) framework[4] for utility and fairness.
  - Utility metrics: F1-Macro, MCC[5], G-Mean
  - Fairness metrics: Demographic Parity Difference (DPD)[6] and Equalized odds (EOD)[7]
    - Fairlearn** to calculate the metrics.

Results

Fidelity

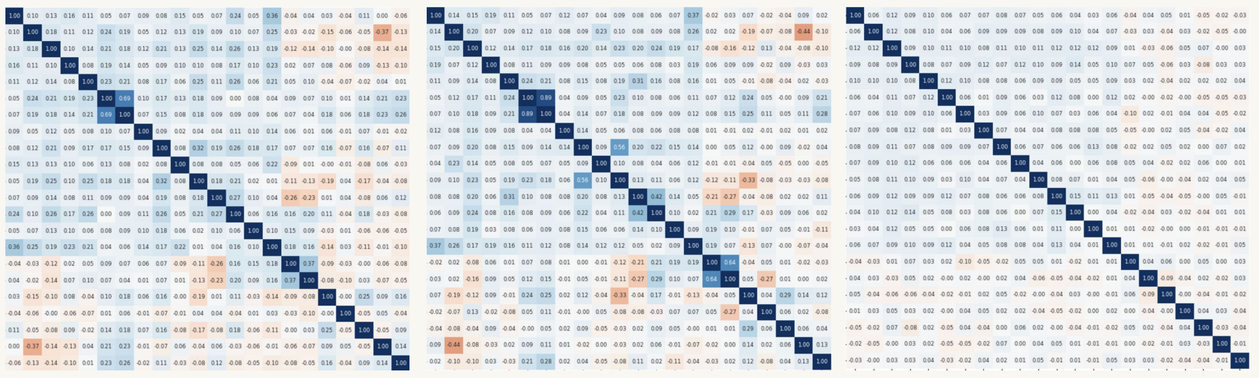


Figure 2: Inter-variable correlations in real data (left), config 15 (middle), config 8 (right)

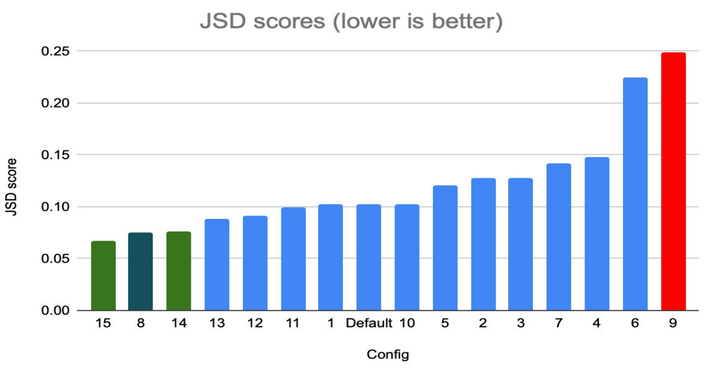


Figure 3: JSD scores for all configurations

Config 15 had the lowest score(best) and config 9 had the highest score(worst). Config 8 achieved a good score but did not preserve inter-variable relationships (Fig 2).

Utility

Config		F1-Macro	MCC[5]	G-Mean
TRTR	RF	0.6530	0.3344	0.5874
	LDA	**0.7384**	**0.4864**	**0.7036**
TSTR_14	RF	**0.6573**	**0.3465**	**0.5897**
	LDA	0.6273	0.2682	0.5732
TSTR_15	RF	0.6066	0.2441	0.5296
	LDA	**0.6174**	**0.2561**	**0.5523**

Table 1: Utility results for best configuration using the TRTR/TSTR framework.

Config 14 achieved the best RF score (classifier of choice), marginally exceeding TRTR baseline. TRTR LDA scores were not improved upon by any classifier.

Fairness

Config	DPD[6]	EOD[7]
TRTR (RF, LDA)	(0.4050, **0.2467**)	(0.4231, **0.2857**)
TSTR_14 (RF, LDA)	(0.1042, **0.0138**)	(0.1346, **0.0096**)
TSTR_15 (RF, LDA)	(0.2120, **0.1384**)	(0.2115, **0.1746**)

Table 2: Fairness results for best configuration using the TRTR/TSTR framework.

TSTR configurations showed better fairness compared to TRTR, with config 14 having the best score.

Config 14 achieves best utility and fairness scores but not the best fidelity

Conclusion

Configurations with high fidelity did not guarantee better downstream utility or fairness. These findings suggest that evaluating synthetic data on fidelity alone is not enough, especially in high-stakes domains like finance where decisions can have lasting economic consequences.

References

*SDV library: https://docs.sdv.dev/sdv
** Fairlean: https://fairlearn.org/
[1] Assefa et al. (2020) - Generating synthetic data in Finance: opportunities, challenges and pitfalls
[2] Xu et al. (2019) - Modelling Tabular Data using Confitional GAN
[3] Hofmann (1994) - German Credit Dataset
[4] Jordon et al. (2018) - Measuring the quality of synthetic data for use in competitions
[5] Chicco & Jurman (2020) - The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation
[6] Dwork et al. (2012) - Fairness through awareness.
[7] Hardt et al. (2016) - Equality of Opportunity in Supervised learning